Computer Network

A computer network is a system in which multiple computers are connected so that they can share information and resources.

Advantages：

-data sharing

-resource sharing: printer、storage、software databases

-communication

Network Software: communication、routing、resource management、security

Network-wide Operating System: distributed systems、cloud computing

4.1 Network Fundamentals

Network Classifications

1)　geographical size

类型	全称	覆盖范围	例子
PAN	Personal Area Network	几米	蓝牙设备
LAN	Local Area Network	一栋楼	学校 / 公司
MAN	Metropolitan Area Network	一个城市	城市网络
WAN	Wide Area Network	大范围	Internet

PAN（Personal Area Network） short-range communication

wireless headset ↔ smartphone

wireless mouse ↔ PC

Bluetooth devices

LAN（Local Area Network） single building or campus

university campus network

company office network

factory network

MAN（Metropolitan Area Network） spanning a local community

city government network

city internet infrastructure

WAN（Wide Area Network） long-distance communication

corporate international network

Internet

2）Ownership

Open Network based on publicly available standards

Example: Internet

because internet use TCP/IP protocol suite

Closed Network（Proprietary Network） owned by a specific organization

Novell Inc.

3) Topology

Bus Topology

In a bus topology, all computers are connected to a shared communication line, called a bus.

Key feature: all machines share the same communication medium

The bus topology became popular with the development of: **Ethernet**

Ethernet is one of the most widely used networking technologies.

Star Topology

In a star topology, all machines are connected to a central node. The central machine coordinates communication.

In wireless networks, the central node is called an: **Access Point (AP)** Example: WiFi router

Hub

A **hub** is a device that broadcasts incoming signals to all connected machines.

Important point: A hub-based network looks like a star, but operates like a bus network. Because the hub simply relays signals to all devices.

Protocols

A **protocol** is a set of rules that governs communication between devices in a network.

protocol = communication rules

Protocol Standards -> because need compatible to communicate with different vendors.

CSMA/CD Carrier Sense Multiple Access with Collision Detection

For Ethernet bus networks

Step 1: listen to the channel-> channel silent -> send

Step 2: monitor the channel while transmitting

Step 3: if collision

Step 4: stop -> waiting for random time ->resend

CSMA/CD is not compatible with wireless star networks in which all machines communicate through a central AP

Hidden Terminal Problem

The hidden terminal problem occurs when two machines cannot detect each other's transmission but both communicate with the same access point.

CSMA/CA Collision Avoidance

Step 1: listen to the channel

Step 2: channel silent -> wait for a short time

Step 3: still silent -> start sending

Step 4: if not silent -> stop -> waiting for random time ->resend

IEEE 802.11（WiFi 标准） wireless networking standards

IEEE（Institute of Electrical and Electronics Engineers）

RTS / ACK

Machine -> Request

AP -> acknowledgment

Machine -> data

Methods of Process Communication

Why combine networks? Sometimes it is necessary to connect existing networks to form an extended communication system.

Repeater: A repeater is a device that simply passes signals between two network segments.

forward signals、amplify signals、without considering the meaning of the signals

Bridge: A bridge connects two networks and forwards messages based on their destination addresses.

Read destination address
If destination is in other networks -> forward
If destination is in this network -> do not forward
reduce network traffic
Switch: A switch is essentially a bridge with multiple connections.
multiport bridge

```
          Computer
              |
Computer-Switch-Computer
              |
          Computer
```

Find destination address -> send to the right port
reduce unnecessary traffic, improve network efficiency
repeater bridge switch The entire system operates as a single large network using the same protocols.
Problem: WiFi network Ethernet network using different protocols.
internet: An internet is a network of interconnected networks.
Router: A router is a device that forwards messages between networks.

Device	Function
repeater	signal relay
bridge	connect 2 LANs
switch	connect many LAN segments
router	connect different networks

depend on internet address (Forwarding Table) -> packet forwarding
Addressing:
local address: MAC address
internet address: IP address
Gateway: A gateway is a point where a network connects to the outside internet.
Router + Access Point

Process Communication

Communication between processes is called **interprocess communication (IPC)**.
Client / Server Model: The client/server model defines processes as either clients or servers**.**
Client → request → Server
Client ← response ← Server
Example 1：Print Server: A printer connected to the network serves as a print server.
Example 2：File Server: A machine with large storage acts as a file server.
Server: execute continuously (Web server\ Email server\ Database server)
Peer-to-Peer Model (P2P): Processes provide services to and receive services from each other.
Peer ↔ Peer
Peer ↔ Peer
Peer ↔ Peer

Example 1：Instant Messaging (Telegram)
Example 2：Online Games (interactive games)
One peer may receive a file and then distribute it to other peers.
Swarm: A collection of peers distributing a file.

Feature	**Client/Server**	**Peer-to-Peer**
structure	centralized	decentralized
server	yes	no
role	fixed roles	equal peers
runtime	server always running	temporary processes

Client/Server
One server provides services to many clients.
P2P
Peers both provide and receive services.
P2P is more efficient. It distributes the service task over many peers.
But it has law problems. Because lack of a centralized server
Peer-to-peer refers to a **communication model**, not a network type.
X Peer-to-Peer Network

Distributed Systems: A system in which software units run as processes on different computers and cooperate through a network.
global search engine\company accounting system\online games\network infrastructure software
Distributed Infrastructure: Early distributed systems were developed independently from scratch.
Now research is revealing a common infrastructure (communication systems\ security systems)
So prefabricated systems exist.
Cluster Computing: Many independent computers work closely together.
High availability- At least one machine will respond.
Load balancing- Workload can be shifted automatically between machines.
Grid Computing: distributed systems that are more loosely coupled than clusters
computers can be anywhere\even personal computers
Volunteer Computing: volunteer computing power
Cloud Computing: huge pools of shared computers allocated to clients as needed
Cloud Computing Advantages: Reliability Scalability
Cloud Computing Concerns: Privacy Security

1. **What is an open network?**

An **open network** is a network based on **publicly available standards and protocols**, allowing products from different vendors to communicate with each other.

2. **Summarize the distinction between a bridge and a switch.**

A bridge connects two networks and forwards messages based on their destination addresses. A switch is essentially a bridge with multiple connections.

3. **What is a router?**

A router is a device that forwards messages between networks based on their IP addresses.

4. **Identify some relationships in society that conform to the client/server model.**

customers and restaurants
students and libraries
users and information services

5. **Identify some protocols used in society.**

traffic rules/postal regulations/telephone communication procedures

6. **Summarize the distinction between cluster computing and grid computing.**

Cluster Computing: Many independent computers work closely together.
Grid Computing: distributed systems that are more loosely coupled than clusters

4.2 The Internet

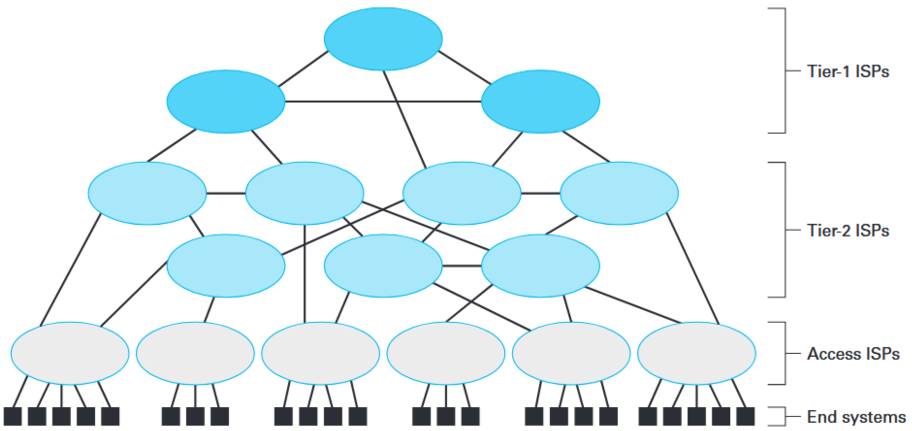
The **Internet** is a worldwide **collection of interconnected networks** that allows millions of computers to communicate with each other.

Internet Architecture

The Internet is built from networks operated by **Internet Service Providers (ISPs)**.

An ISP is an organization that constructs and maintains a network and provides Internet access.

ISP hierarchy



Tier-1 ISPs

very high-speed international WANs

form the backbone of the Internet

operated by large communications companies

Tier-2 ISPs

connect to tier-1 ISPs

more regional

lower capacity

Tier-3 ISPs (Access ISPs)

Provide Internet access to homes and businesses.

Examples: cable companies/telephone companies/universities

End Systems / Hosts

Devices connected to the Internet are called end systems or hosts.

Examples: PCs/laptops/smartphones/cameras/cars/home appliances

Wireless Internet and Hot Spots

A **wireless access point (AP)** connects to an **access ISP** and provides Internet access.

The area covered by an AP is called a **hot spot**.

Hot spots exist in: homes/offices/hotels/parks/cities

Cellular network analogy: Cellular systems also create **cells**, which function similarly to hot spots.

Internet2

Internet2 is a **research network** used by universities, industry, and government.

Purpose: research high-bandwidth applications

Examples: remote robotic surgery

The Last Mile Problem

The **last mile problem** refers to the difficulty and cost of replacing the old analog lines that connect homes to modern digital networks.

Many modern networks use **high-speed digital technology** like **fiber optics**. However, connections from the network to individual homes often use **old analog infrastructure**. This issue is called the **last mile problem**.

Solutions: DSL modem/cable modem/satellite uplink/fiber-optic connections

Internet Addressing

IP Address

An IP address is a unique identifying address assigned to each computer on the Internet.

IP = Internet Protocol

Address length: IPv4 → 32 bits/IPv6 → 128 bits

Address allocation

ICANN (Internet Corporation for Assigned Names and Numbers)

ICANN
↓
ISP
↓
individual machines

Dotted Decimal Notation

32 bits
→ 4 bytes
→ each byte written in decimal

Domain Names

domain name: mu.edu

Domain: administrative region (university/company/government agency/club)

Top-Level Domain（TLD）

mu.edu
↑
TLD

TLD	**meaning**
com	commercial
edu	educational
gov	government
org	nonprofit
museum	museums
info	unrestricted
net	originally for ISPs

Country-code TLD
au → Australia
ca → Canada
jp → Japan
Subdomains
eagle.mu.edu
host.domain.TLD
overthruster.propulsion.yoyodyne.com
host.subdomain.domain.TLD
DNS（Domain Name System）
The **Domain Name System (DNS)** is an Internet-wide directory system that translates domain names into IP addresses.
domain name
↓
DNS lookup
↓
IP address
DNS Lookup
user enters domain name -> request sent to name server -> server returns IP address
Domains without own infrastructure
kingsandqueens.org
he organization can contract with an access ISP.
The ISP will: help register the domain name. Include the domain in the ISP's name server.

Internet Applications
In the early days of the Internet, most applications were separate and simple programs.
Each program followed its own network protocol.
NNTP (Network News Transfer Protocol)
Used by: newsreader applications
Purpose: read and distribute network news
FTP (File Transfer Protocol)
Purpose: listing files/copying files across the network
Telnet
Purpose: remote login/access another computer remotely
SSH (Secure Shell)
Purpose: secure remote access
Modern Internet Applications: Many traditional applications are now handled through webpages. **HTTP (Hyper Text Transfer Protocol)**
Purpose: web communication
Electronic Mail
SMTP (Simple Mail Transfer Protocol) SMTP is the protocol used to transfer email messages between mail servers.
SMTP transmission process: sending mail server contacts receiving mail server -> DNS is used to find the IP address of the destination mail server -> servers communicate using SMTP

commands.

```
220 mail.tardis.edu SMTP Sendmail Gallifrey-1.0; Fri, 23 Aug 2413 14:34:10
HELO mail.skaro.gov
250 mail.tardis.edu Hello mail.skaro.gov, pleased to meet you
MAIL From: dalek@skaro.gov
250 2.1.0 dalek@skaro.gov... Sender ok
RCPT To: doctor@tardis.edu
250 2.1.5 doctor@tardis.edu... Recipient ok
DATA
354 Enter mail, end with "." on a line by itself
Subject: Extermination.

EXTERMINATE!
Regards, Dalek
.
250 2.0.0 r7NJYAE1028071 Message accepted for delivery
QUIT
221 2.0.0 mail.tardis.edu closing connection
```

Email Access Protocols

POP3 (Post Office Protocol version 3)

download emails to local computer

emails stored on user's machine

simple

IMAP (Internet Mail Access Protocol)

email stored on mail server

user can access email from multiple computers

VoIP

VoIP (Voice over Internet Protocol) is a technology that uses the **Internet infrastructure** to provide **voice communication** similar to traditional telephone systems.

Basic working principle:

Two processes on different machines

↓

transfer audio data

↓

via the P2P model

Why VoIP is not so simple in reality? initiating calls/linking with traditional telephone systems/emergency services (911)

VoIP Soft Phones

Example: Skype/Zoom

easy to use/no special hardware

proprietary system

PC

↓

Internet

↓

PC

Analog Telephone Adapter (ATA) connect traditional phone to Internet

old phone

↓
adapter
↓
Internet
Embedded VoIP Phones dedicated IP telephone
VoIP over Ethernet
cost reduction/more features
IP phone
↓
Ethernet
↓
Internet
Smartphone VoIP mobile VoIP over IP networks
4G network
↓
IP network
↓
Internet
smartphone = Internet host
The Generations of Wireless Telephones
1G = analog voice
2G = digital voice + SMS
3G = mobile internet
4G = high-speed IP network
Internet Multimedia Streaming
An enormous portion of current Internet traffic is used for transporting audio and video across the Internet in real-time, known as **streaming**.
N-unicast (1 sender → N receivers)

```
Server
   │
   ├──── Client A
   ├──── Client B
   ├──── Client C
   └──── Client D
```

P2P Streaming
Example: BitTorrent
Server → Peer1
Peer1 → Peer2
Peer1 → Peer3
Peer2 → Peer4
Peer3 → Peer5
Multicast
1 message → many receivers
Server → Router

```
Router
 ├──── Client A
 ├──── Client B
 └──── Client C
```

CDN

```
      Netflix Server (USA)
                 │
  ┌──────────────┴──────────────┐
Japan CDN                  Europe CDN
  │                             │
Users                         Users
```

User → ISP → CDN server

Anycast: find the closest server

```
Streaming Problem
        │
        ▼
   N-unicast
        │
   (server overload)
        │
  ┌─────┴──────┐
  ▼            ▼
P2P        Multicast
               │
     (router support needed)
               │
               ▼
      On-demand streaming
               │
               ▼
              CDN
               │
               ▼
            Anycast
```

1. **What is the purpose of tier-1 and tier-2 ISPs? What is the purpose of access ISPs?**

Tier-1 ISPs form the backbone of the Internet and connect directly to each other.

Tier-2 ISPs connect to tier-1 ISPs and provide service to smaller networks and access ISPs.

Access ISPs connect end users and their devices to the Internet.

2. **What is DNS?**

Domain name system. It translates domain names into IP addresses.

3. **What bit pattern is represented by 3.6.9 in dotted decimal notation? Express the bit pattern 0001010100011100 using dotted decimal notation.**

3.6.9 -> 00110000 01100000 10010000

00010101 00011100 -> 21.28

4. **In what way is the structure of a mnemonic address of a computer on the Internet (such as overthruster.propulsion.yoyodyne.com) similar to a traditional postal address? Does this same structure occur in IP addresses?**

overthruster.propulsion.yoyodyne.com -> room.street.city.country

No.

5. Name three types of servers found on the Internet and tell what each does.

Web server – provides web pages to clients.

Mail server – sends and receives electronic mail.

File server – stores and distributes files.

6. What aspects of network communication are described by a protocol?

message format

message order

actions taken on transmission and receipt of messages

7. In what way do the P2P and multicast approaches to Internet radio broadcast differ from N-unicast?

N-unicast (1 sender → N receivers)

```
Server
   │
   ├──── Client A
   ├──── Client B
   ├──── Client C
   └──── Client D
```

P2P Streaming

Example: BitTorrent

Server → Peer1

Peer1 → Peer2

Peer1 → Peer3

Peer2 → Peer4

Peer3 → Peer5

Multicast

1 message → many receivers

Server → Router

```
Router
  ├──── Client A
  ├──── Client B
  └──── Client C
```

8. What criteria should one consider when choosing one of the four types of VoIP?

existing equipment/cost/mobility/network availability

4.3 The World Wide Web

Hypertext a document that contains links to other documents

Hyperlinks a reference in a hypertext document that points to another document or resource

Hypertext (document)

└──── Hyperlinks (connections)

Web Implementation

Browser A browser resides on the user's computer and retrieves and displays web documents.

obtain requested materials/present them to the user

browser = client

Webserver A webserver provides access to hypertext documents requested by browsers.

store web documents/send them to browsers

Browser (client) → request

Webserver → response

HTTP (Hypertext Transfer Protocol) HTTP is the protocol used to transfer hypertext between browsers and webservers.

URL（Uniform Resource Locator） A URL is a unique address used to locate a document on the Web.

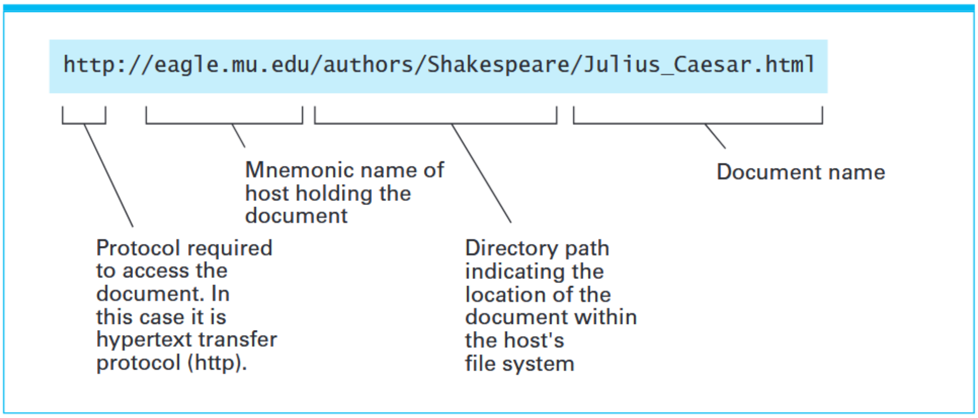
1 user enters URL

2 browser contacts server

3 server returns document

4 browser displays webpage

Figure 4.8 A typical URL



默认 protocol = HTTP

www.google.com -> http://www.google.com

HTML

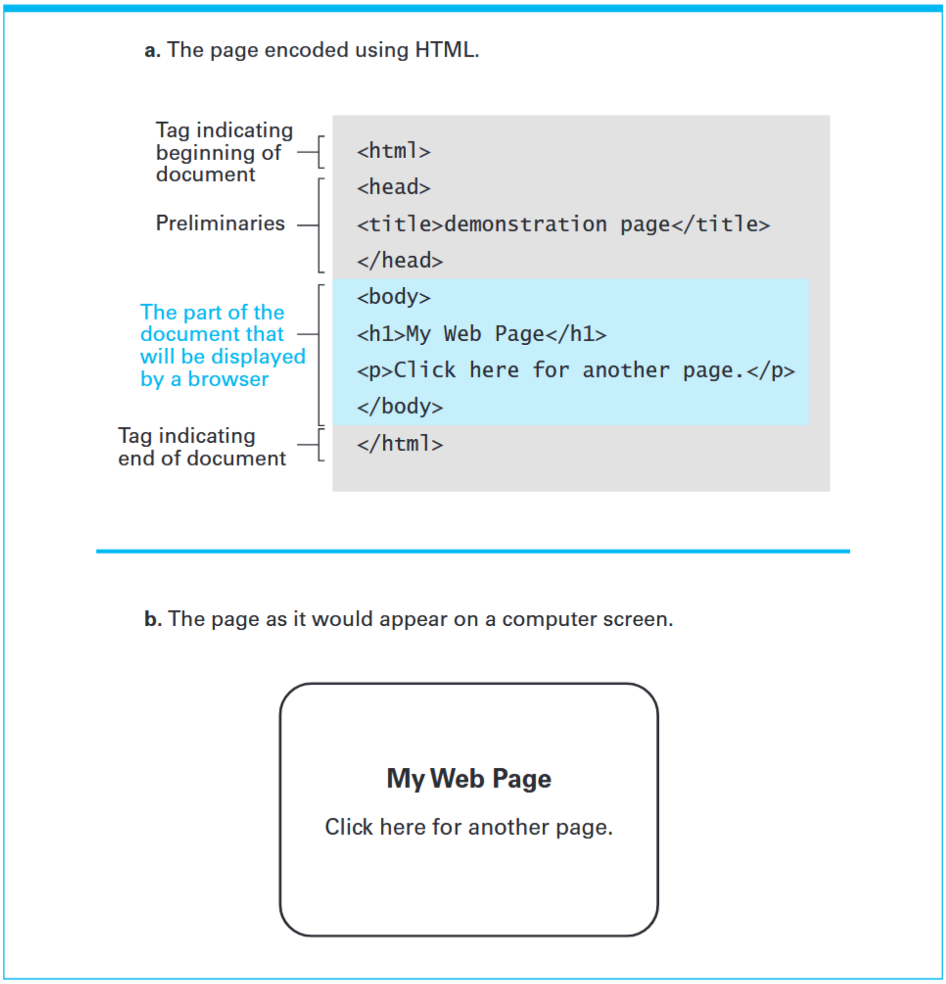
Hypertext Markup Language (HTML)

HTML is a system of tags used to describe how a webpage should be displayed.

HTML document = text + **tags**

Tag Tags describe how the document should appear on the screen.

Figure 4.9 A simple webpage



Head

<head>

<title>My Web Page</title>

</head>

Body

<body>

</body>

Heading tag

<h1>My Web Page</h1> level-one heading

Paragraph tag

<p>Click here for another page.</p>

Hyperlink

here anchor tag

Image tag

<img src="http://Images.com/Images/OurPic.jpg">

img tag

src = source

Figure 4.10 An enhanced simple webpage

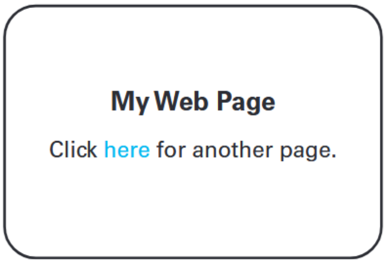
a. The page encoded using HTML.

```
<html>
<head>
<title>demonstration page</title>
</head>
<body>
<h1>My Web Page</h1>
<p>Click
   <a href="http://crafty.com/demo.html">
   here
   </a>
   for another page.</p>
</body>
</html>
```

Anchor tag containing parameter

Closing anchor tag

b. The page as it would appear on a computer screen.



XML

eXtensible Markup Language (XML) XML is a standardized style for designing markup languages used to represent data as text files.

HTML 是一种 **markup language** how something looks

XML 是 markup language 的设计标准 what something means（Semantics 语义）

HTML 不完全符合 XML

XHTML: HTML version that strictly follows XML rules

Example:

```
<staff clef = "treble"> <key>C minor</key>
<time> 2/4 </time>
<measure> <rest> egth </rest> <notes> egth G,
egth G, egth G </notes></measure>
<measure> <notes> hlf E </notes></measure>
</staff>
```

semantic tags: tags describing meaning
search engines (websites that assist users in locating Web material pertaining to a subject of interest)
better search
search: recipes without spinach
search engines identify <ingredient> -> "This recipe does not contain spinach"
Client-Side and Server-Side Activities
User → Browser → Webserver ← HTML page
Browser → display page
hyperlink:
Browser → request new page
Server → send page
Client-side Activities: Activities performed by the client (browser).
choose destination/enter travel date
Server-side Activities: Activities performed by the server.
create customized webpage (webpage generated based on user input)
client request → server processing → server response
Client-side Tech: Java applets/ Adobe Flash
Server-side Tech:
CGI (Common Gateway Interface): execute programs on server
Java servlets/ JavaServer Pages (JSP)/ Active Server Pages (ASP)/ PHP

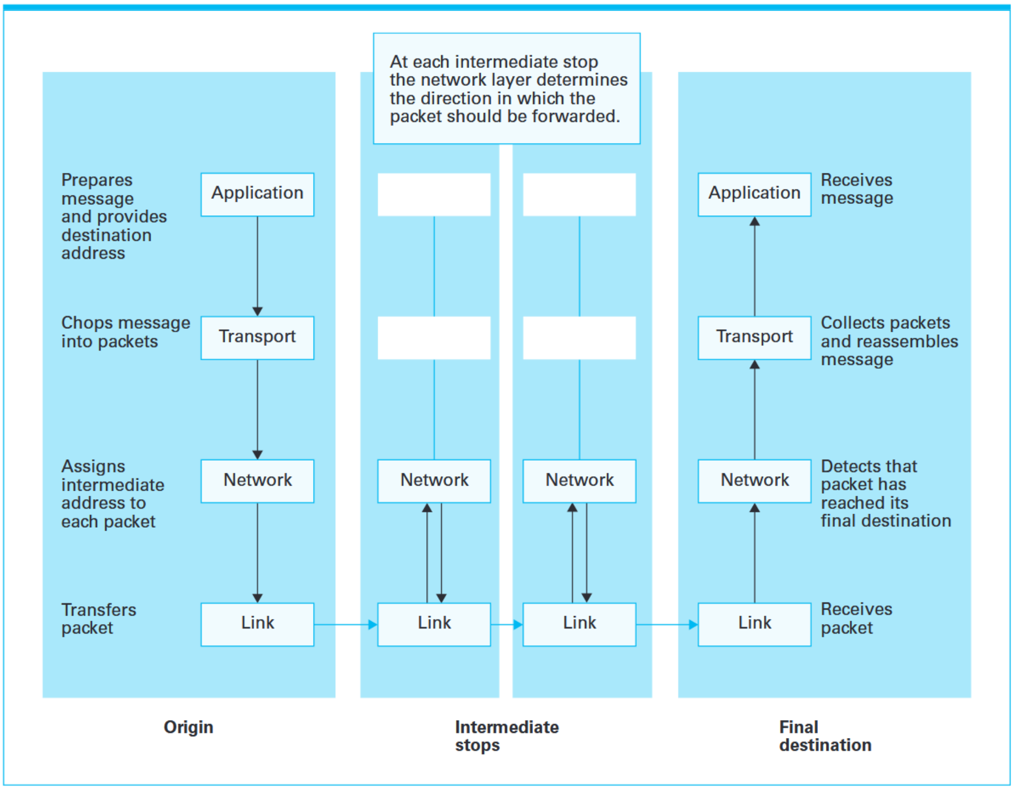
1. What is a URL? What is a browser?
URL（Uniform Resource Locator）A URL is a unique address used to locate a document on the Web.
A browser resides on the user's computer and retrieves and displays web documents.
2. What is a markup language?
A markup language is a system of tags used to encode documents as text files.
3. What is the difference between HTML and XML?
HTML focus on appearance
XML focus on semantics (meaning) or structure
4. What is the purpose of each of the following HTML tags?
**a. <html>　b. <head>　c. </p>　d. **
a. <html>Indicates the beginning of an HTML document.
b. <head>Contains preliminary information about the document (such as the title).
c. </p>Indicates the end of a paragraph.
**d. **Indicates the end of an anchor (hyperlink).
5. To what do the terms client side and server side refer?
Client-side Activities: Activities performed by the client (browser).
Server-side Activities: Activities performed by the server to process client requests and construct webpages.

4.4 Internet Protocols

The Layered Approach to Internet Software

The Internet software is organized into layers. Each layer provides services to the layer above it and uses services from the layer below it.

Figure 4.14 Following a message through the Internet



application layer

This is where messages originate.

Includes programs like web browsers, FTP clients, SSH, etc.

Responsible for providing addresses (via DNS) compatible with the Internet.

transport layer

Takes messages from the application layer and splits them into smaller segments for easier handling.

Adds sequence numbers so the original message can be reassembled at the destination.

Passes segments to the network layer.

network layer

Decides the path each packet takes through the Internet.

Maintains forwarding tables in routers to route packets correctly.

Packets may take different paths and arrive out of order.

link layer

Handles the actual transmission of packets between adjacent machines/networks.

Deals with network-specific details like Ethernet or WiFi protocols (CSMA/CD, CSMA/CA).

At intermediate stops (routers), only link + network layers are active.

Encapsulation

Each layer wraps the message in its own header (like nested envelopes).

Each layer focuses only on its responsibilities, ignoring details of other layers.

Packets： Packet reordering: Because packets can take different paths, sequence numbers ensure proper assembly.

port numbers: Used by transport layer to deliver messages to the correct application unit.

Layer	Data Unit Name	Header (added info)	Trailer (added info)	中文解释
Application	Message / Data	Depends on protocol (e.g., HTTP headers, FTP commands)	Optional	应用层数据，包含协议相关信息（例如 HTTP 头部）
Transport	Segment (TCP) / Datagram (UDP)	**Port numbers** (source & destination), **Sequence number**, **Checksum**	Optional (TCP may include trailer for checksum)	传输层把消息拆成段，添加端口号、序号、校验和等信息
Network	Packet	**Source & destination IP addresses**, TTL (time-to-live), protocol info	Optional (some encapsulation info)	网络层称为包，添加 IP 地址、TTL 等，用于路由
Link	Frame	**MAC addresses** (source & destination), Frame type	**Error detection (CRC)**	链路层称为帧，加上物理地址和错误检测字段

Application layer creates a message → hands to transport layer.

Transport layer splits into packets, adds sequence & port numbers → hands to network layer.

Network layer decides path for each packet → hands to link layer for actual sending.

Link layer sends packet to next router or destination machine.

At intermediate routers, only network + link layers operate:

Network layer checks forwarding table.

Link layer transmits to next hop.

At destination machine:

Link layer receives packets → network layer checks destination → transport layer reassembles segments → application layer receives original message.

The TCP/IP Protocol Suite

The TCP/IP suite is the set of protocols used to implement the four-layer Internet model

.

Open System Interconnection (OSI) OSI Model: A 7-layer reference model by ISO.

Transmission Control Protocol (TCP)

Connection-oriented: Before sending the message, TCP first establishes a connection with the destination.

Reliable: Uses acknowledgments and retransmissions to ensure all packets arrive.

Flow control: Adjust sending rate so the receiver isn't overwhelmed.

Congestion control: Adjust sending rate to prevent network congestion.

Slower and more complex than UDP but ensures delivery correctness.

User Datagram Protocol (UDP)

Connectionless: Just sends the packet and forgets about it.

Unreliable: No guarantees of delivery or order.

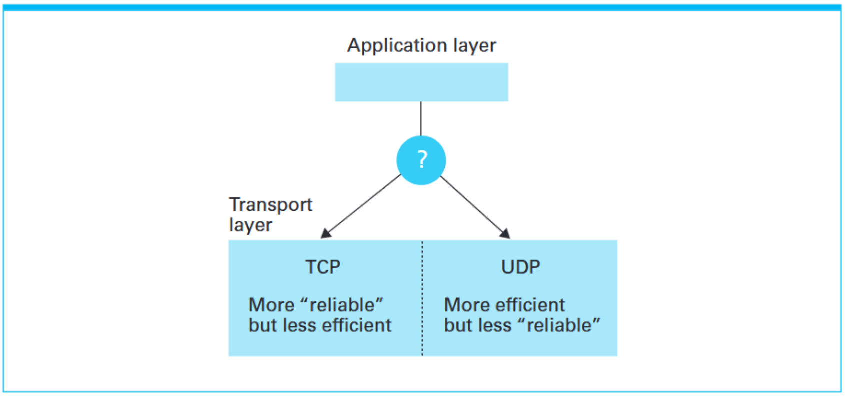
Efficient: Less overhead → faster for time-sensitive tasks.

Common uses:

DNS lookups (fast queries)

VoIP / video calls (low latency matters more than perfect reliability)

Figure 4.15 Choosing between TCP and UDP



Internet Protocol (IP)

The network layer is implemented by IP (Internet Protocol):

Forwarding: Send packets to the next hop toward the destination.

Routing: Update forwarding tables based on network conditions.

Example: Router fails → traffic rerouted

Network congestion → packets rerouted

Hop count/Time To Live (TTL): An interesting feature associated with forwarding is that each time an IP network layer at a message's origin prepares a packet, it appends a value called a hop count, or time to live, to that packet.

Each packet gets a TTL value (e.g., 64).

Every time a packet passes a router, TTL decreases by 1.

TTL prevents packets from circulating forever.

IP Versions

IPv4

32-bit addresses

Still widely used, but limited in address space

IPv6

128-bit addresses

Solves address shortage

Supports multicast and other features

Gradually replacing IPv4

Choosing TCP or UDP

Depends on quality-of-service needs:

Need reliability → TCP

Need speed/efficiency → UDP

Layer	Protocol	Key Feature	Example Use	中文说明
Transport	TCP	Reliable, connection-oriented, flow & congestion control	Email, HTTP download	可靠，慢，保证传输
Transport	UDP	Connectionless, fast, minimal overhead	DNS, VoIP	快，不保证到达
Network	IP	Routing & forwarding, TTL, addressing	Internet packet delivery	决定路线，防止循环

1. What layers of the Internet software hierarchy are not needed at a router?

Transportation layer. Application layer. used only at the source and destination computers.

2. What are some differences between a transport layer based on the TCP protocol and another based on the UDP protocol?

TCP: Reliable, connection-oriented, flow & congestion control.

UDP: Connectionless, fast, minimal overhead

3. How does the transport layer determine which unit with the application layer should receive an incoming message?

The transport layer uses port numbers to deliver messages to the correct application.

Each application process is assigned a unique port number, and the transport layer appends this port number to the message.

At the destination, the transport layer reads the port number and forwards the message to the correct application.

4. What keeps a computer on the Internet from recording copies of all the messages passing through it?

Only the Link layer and Network layer at intermediate routers see the packets.

Routers forward packets based on destination addresses, not on the application message content. (only need the information in the header)

The Transport layer and Application layer (where messages are assembled and interpreted) exist only on the source and destination computers.

4.5 Security

Forms of Attack

Malware (恶意软件)**:** Malicious software (malware) is software designed to damage systems, steal information, or disrupt operations.

Virus: A virus is software that infects a computer by attaching itself to existing programs.

-Key characteristics: Needs a host program/ Executes when the host program runs/ Can replicate by infecting other programs

-Possible damage: degrade the operating system/ erase storage/ corrupt programs and data

Worm: A worm is an autonomous program that spreads through a network.

-Key characteristics: Self-replicating(复制)/ Spreads automatically across computers/ Does not need a host program

-Main danger: create massive numbers of copies -> slow down systems/ overload networks

Trojan horse: A Trojan horse is a program that disguises(伪装) itself as useful software.

-Key idea: User voluntarily installs it, believing it is safe. -> steal data/ damage files/ open backdoors

Sometimes it stays dormant (潜伏) until triggered by: a specific date/ a specific event

Spyware (sniffing software**)** Spyware collects information about a user's activities and sends it back to the attacker.

What it can collect: browsing activity/ personal information/ keystrokes

Phishing (🎣)**:** Phishing is a technique of tricking users into voluntarily giving sensitive information.

Common method: Fake emails pretending to be: banks/ government agencies/ law enforcement

denial of service (DoS): A Denial of Service (DoS) attack overloads a system with huge numbers of requests. (拒绝服务攻击)

The server becomes: extremely slow/ completely unavailable. Companies' websites may be shut down temporarily.

botnet (僵尸网络): A botnet is a network of infected computers controlled by an attacker.

Process: attacker installs malware on many computers -> waits for command -> all machines send messages to the target simultaneously

This creates a massive traffic flood.

spam (垃圾邮件): Spam is unwanted mass email.

Unlike DoS attacks, spam usually does not crash systems.

overwhelms users/ spreads phishing/ distributes Trojan horses and viruses.

Protection and Cures

It is better to prevent attacks than to fix the damage afterward.

Firewall: A firewall is a program that filters network traffic entering or leaving a network or computer.

It is often installed at: the gateway of an intranet(企业内部互联网)/ an individual computer

Main Functions：Block suspicious traffic/ Stop Denial of Service attacks/ Prevent **spoofing:** Spoofing means pretending to be another computer or user.

spam filters: Spam filters are specialized firewalls designed to block unwanted email.

Many spam filters use machine learning techniques.

User labels emails as spam -> System learns patterns -> Filter automatically blocks future spam

proxy server (代理服务器)**:** A proxy server acts as an intermediary between a client and a server.

Client → Proxy Server → Server

Server thinks the proxy is the real client.

The real client's identity is hidden. / The proxy server can check incoming data.

Auditing Software: detect problems before they become serious

antivirus software: Antivirus software detects and removes malware such as: viruses/ spyware/ other infections

New malware appears constantly. Therefore antivirus software must be **regularly updated**.

Safe User Behavior: Never: open unknown email attachments/ download untrusted software/ click suspicious pop-up ads

Encryption

Encryption = converting readable data into encoded form so unauthorized users cannot understand it.

HTTPS (Hypertext Transfer Protocol Secure) is the secure version of HTTP.

Secure Sockets Layer (SSL)

Secure Sockets Layer (SSL) is a protocol that provides secure communication between a web client and server.

It forms the backbone of HTTPS.

encrypt transmitted data/ protect user privacy/ prevent eavesdropping (偷听)

登录银行 加密连接 保证网络传输过程中别人看不懂数据

publickey encryption: One of the most important ideas in modern cryptography is public-key encryption.

Two different keys are used.

knowing how encryption works does NOT allow someone to decrypt messages.

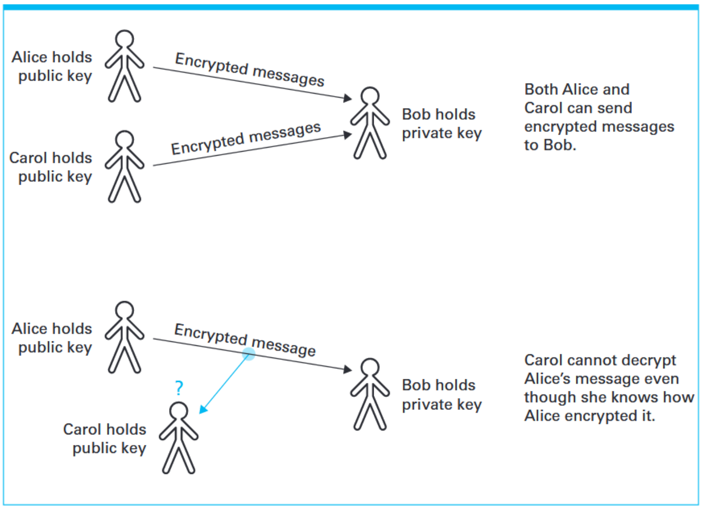
A **public-key encryption system** uses two keys.

public key: used to encrypt messages/ can be shared with everyone

private key: used to decrypt messages/ kept secret by the receiver

银行会：公开 public key 。浏览器用 public key 加密数据发给银行。只有银行有 private key 所以只有银行能解密。

Figure 4.16 Public key encryption



Problem: Fake Public Keys (公钥伪造问题) 如何确定公钥是真的?
Spoofing (身份伪造).If users encrypt data with the fake key, the attacker could read everything.
certificate authorities (CA)
Certificate authorities (CA) are trusted organizations that verify identities and manage public keys.
Certificates
A certificate is a package containing: a person or organization's name and their public key
When a browser connects to a secure site: The server sends its certificate. -> The browser verifies it using the certificate authority. -> If valid, the secure connection begins.
浏览器检查证书 确定网站是否是真的
Authentication
Authentication means verifying that the sender of a message really is who they claim to be.
digital signature
Public-key systems can also create digital signatures.
Sometimes the encryption process is reversed: The sender encrypts data using their private key.
Sender encrypts the message (or a hash of it) with their private key. -> This encrypted data becomes the digital signature. -> Receiver uses the sender's public key to decrypt it.
银行 private key 生成 digital signature。发送：message + signature。浏览器收到后：用银行 public key 验证。如果成功：确认是银行发的内容没有被修改
Legal Approaches to Network Security
Another approach to improving network security is to apply legal remedies. Governments create laws that define illegal activities such as hacking, malware distribution, or privacy violations.
Two obstacles of legal solutions: ① Making something illegal does not stop it.
② The Internet is international.
Computer Fraud and Abuse Act (CFAA) 打击黑客、病毒、非法访问
Electronic Communications Privacy Act (ECPA) 保护通信隐私
Communications Assistance for Law Enforcement Act (CALEA) 支持执法监听
USA PATRIOT Act 扩大反恐监控
Anticybersquatting Consumer Protection Act 防止域名抢注

1. What is phishing? How are computers secured against it?
Phishing is a technique of tricking users into voluntarily giving sensitive information.
-user awareness and caution when responding to emails or websites
-spam filters that detect suspicious messages
-security software that warns users about fraudulent websites
2. What distinction is there between the types of firewalls that can be placed at a domain's gateway as opposed to an individual host within the domain?
Gateway firewall（网关防火墙）
A firewall placed at a domain's gateway filters traffic entering and leaving the entire network. It protects the whole intranet by blocking suspicious messages from outside sources.
Host firewall（主机防火墙）
A firewall installed on an individual host protects a single computer by blocking incoming

messages directed to applications that are not being used on that computer.

3. Technically, the term data refers to representations of information, whereas information refers to the underlying meaning. Does the use of passwords protect data or information? Does the use of encryption protect data or information?

Passwords protect **information**(信息内容) by controlling access to it. They prevent unauthorized users from obtaining the meaning of the data.

Encryption protects **data**(数据本身)by encoding it so that even if the data are intercepted during transmission, the underlying information remains confidential.

4. What advantage does public-key encryption have over more traditional encryption techniques?

Public-key encryption does not require two people to share a secret key beforehand.

The sender can encrypt the message using the receiver's public key, and only the receiver can decrypt it using their private key.

This makes secure communication easier on the Internet.

5. What problems are associated with legal attempts to protect against network security problems?

Legal approaches have two main problems:

1. Declaring an action illegal does not prevent it; it only provides legal recourse after the damage has occurred.
2. Because the Internet is international, laws differ among countries, making enforcement difficult.